



Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping, Paper folding, cutting, and patterns in 2 and 3 dimensions

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	0	1	0	0	1	1	2	1	0	0.75	2
2 Marks Count	1	0	2	1	1	1	0	1	0	0.88	2
Total Marks	2	1	4	2	3	3	2	3	1	2.5	4

10.1

Assembling Pieces (4)

10.1.1 Assembling Pieces: GATE CH 2022 | GA Question: 10



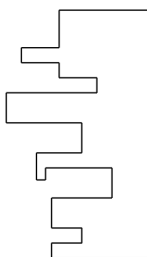
Which one of the following sets of pieces can be assembled to form a square with a single round hole near the center? Pieces cannot overlap.

- A.
- B.
- C.
- D.

gatech-2022 spatial-aptitude assembling-pieces

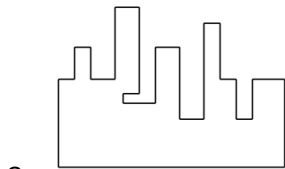
Answer key

10.1.2 Assembling Pieces: GATE CSE 2021 Set 2 | GA Question: 7



A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

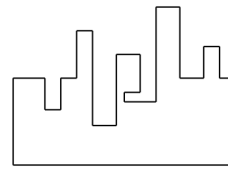
- A.
- B.



C.

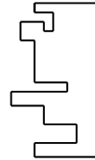
gatecse-2021-set2 spatial-aptitude assembling-pieces two-marks

Answer key

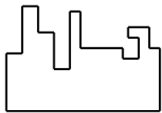


D.

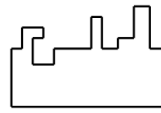
10.1.3 Assembling Pieces: GATE Mechanical 2021 Set 1 | GA Question: 7



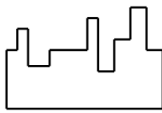
A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.



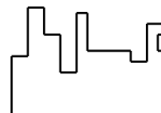
A.



B.



C.

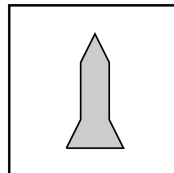


D.

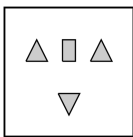
gatecse-2021-set1 spatial-aptitude assembling-pieces

Answer key

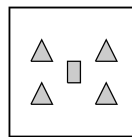
10.1.4 Assembling Pieces: GATE Mechanical 2022 Set 2 | GA Question: 5



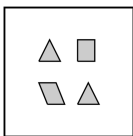
Which one of the groups given below can be assembled to get the shape that is shown above using each piece only once without overlapping with each other? (rotation and translation operations may be used).



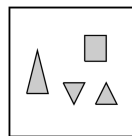
A.



B.



C.



D.

gatecse-2022-set2 spatial-aptitude assembling-pieces

Answer key

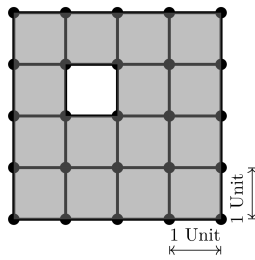
10.2

Counting Figure (1)

10.2.1 Counting Figure: GATE ECE 2022 | GA Question: 4



The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.



What is the maximum number of squares without a “hole in the interior” that can be formed within the 4×4 grid using the unit squares as building blocks?

- A. 15 B. 20 C. 21 D. 26

gateeece-2022 spatial-aptitude patterns-in-two-dimensions counting-figure

Answer key

10.3 Grouping (1)

10.3.1 Grouping: GATE Civil 2024 Set 1 | GA Question: 9



In the 4×4 array shown below, each cell of the first three columns has either a cross (X) or a number, as per the given rule.

1	1	2	
2	X	3	
2	X	4	
1	2	X	

Rule: The number in a cell represents the count of crosses around its immediate neighboring cells (left, right, top, bottom, diagonals).

As per this rule, the **maximum** number of crosses possible in the empty column is

- A. 0 B. 1 C. 2 D. 3

gatecivil-2024-set1 grouping spatial-aptitude

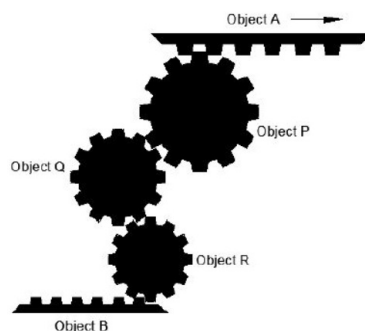
Answer key

10.4 Image Rotation (7)

10.4.1 Image Rotation: GATE 2024 Electrical | GA Question: 9



The assembly shown below has three teathed circular objects (Pinions) and two teathed flat objects (Racks), which are perfectly mating with each other. Pinions can only rotate clockwise or anti-clockwise staying at its own center. Racks can translate towards the left ($\leftarrow$) or the right ($\rightarrow$) direction.



If the object A (Rack) is translating towards the right ($\rightarrow$) direction, the correct statement among the following is

- A. Object **B** translates towards the right direction.
- B. Object **B** translates towards the left direction.
- C. Object **R** rotates in the anticlockwise direction.
- D. Object **Q** rotates in the clockwise direction.

gate2024-ee image-rotation spatial-aptitude

Answer key

10.4.2 Image Rotation: GATE CSE 2022 | GA Question: 5



A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters.



From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.

- A.
- B.
- C.
- D.

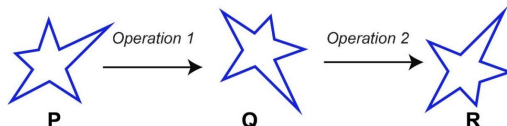
gatecse-2022 spatial-aptitude image-rotation one-mark

Answer key

10.4.3 Image Rotation: GATE CSE 2023 | GA Question: 10



Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?



- A. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- B. Operation 1: A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- C. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line
- D. Operation 1: A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line

gatecse-2023 spatial-aptitude image-rotation two-marks

Answer key

10.4.4 Image Rotation: GATE Civil 2023 Set 2 | GA Question: 5

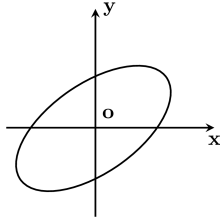
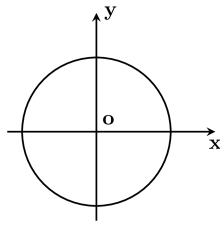


Consider a circle with its centre at the origin (**O**), as shown. Two operations are allowed on the circle.

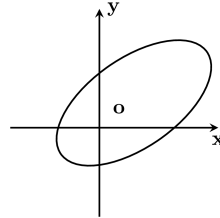
Operation 1 : Scale independently along the x and y axes.

Operation 2 : Rotation in any direction about the origin.

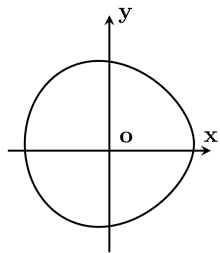
Which figure among the options can be achieved through a combination of these two operations on the given circle?



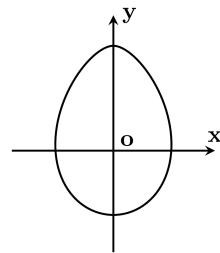
A.



B.



C.



D.

gatecivil-2023-set2 spatial-aptitude image-rotation

Answer key

10.4.5 Image Rotation: GATE Civil 2024 Set 2 | GA Question: 10



Visualize a cube that is held with one of the four body diagonals aligned to the vertical axis. Rotate the cube about this axis such that its view remains unchanged. The magnitude of the minimum angle of rotation is

A. 120°

B. 60°

C. 90°

D. 180°

gatecivil-2024-set2 spatial-aptitude image-rotation

Answer key

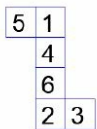
10.4.6 Image Rotation: GATE DS&AI 2024 | GA Question: 9



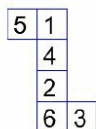
Three different views of a dice are shown in the figure below.



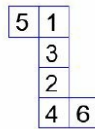
The piece of paper that can be folded to make this dice is



A.



B.



C.



D.

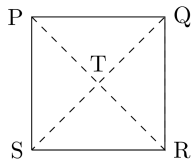
gate-ds-ai-2024 spatial-aptitude image-rotation two-marks

Answer key

10.4.7 Image Rotation: GATE ECE 2022 | GA Question: 10



Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively.



Let X , Y and Z represent the following operations:

X : rotation of the square by 180° with respect to the $S - Q$ axis.

Y : rotation of the square by 180° with respect to the $P - R$ axis.

Z : rotation of the square by 90° clockwise with respect to the axis perpendicular, going into the screen and passing through the point T .

Consider the following three distinct sequences of operation (which are applied in the left to right order).

1. $XYZZ$
2. XY
3. $ZZZZ$

Which one of the following statements is correct as per the information provided above?

- A. The sequence of operations (1) and (2) are equivalent
- B. The sequence of operations (1) and (3) are equivalent
- C. The sequence of operations (2) and (3) are equivalent
- D. The sequence of operations (1), (2) and (3) are equivalent

gatece-2022 spatial-aptitude image-rotation

Answer key

10.5

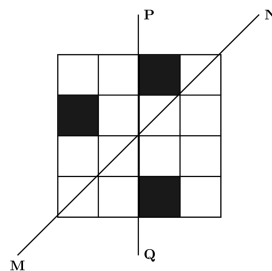
Mirror Image (7)

10.5.1 Mirror Image: GATE CH 2023 | GA Question: 5



A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.

The given figure consists of 16 unit squares arranged as shown. In addition to the three black squares, what is the minimum number of squares that must be coloured black, such that both PQ and MN form lines of symmetry? (The figure is representative)



- A. 3
- B. 4
- C. 5
- D. 6

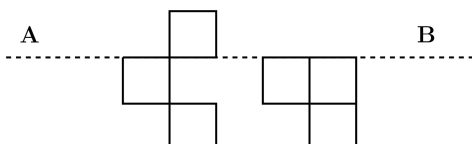
gatech-2023 spatial-aptitude mirror-image

Answer key

10.5.2 Mirror Image: GATE CSE 2024 | Set 1 | GA Question: 10



The least number of squares to be added in the figure to make AB a line of symmetry is



A. 6

B. 4

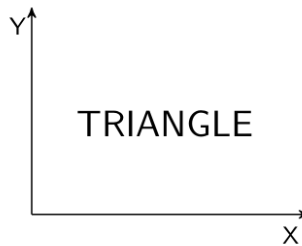
C. 5

D. 7

gatecse2024-set1 spatial-aptitude mirror-image two-marks

Answer key

10.5.3 Mirror Image: GATE Civil 2021 Set 1 | GA Question: 2



The mirror image of the above text about the X -axis is

A. TRIANGLE

B. TRIANGLE

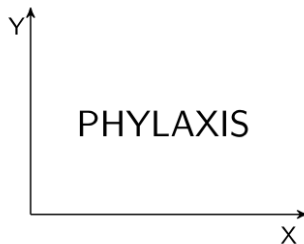
C. TRIANGLE

D. TRIANGLE

gatecivil-2021-set1 mirror-image

Answer key

10.5.4 Mirror Image: GATE Civil 2021 Set 2 | GA Question: 2



The mirror image of the above text about the x -axis is

A. PHYLAXIS

B. PHYLAXIS

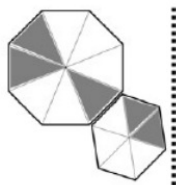
C. PHYLAXIS

D. PHYLAXIS

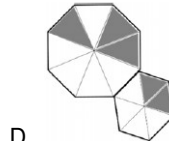
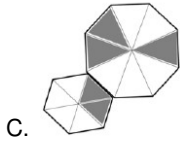
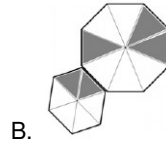
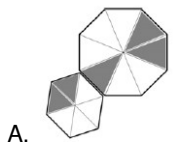
gatecivil-2021-set2 mirror-image

Answer key

10.5.5 Mirror Image: GATE Civil 2022 Set 2 | GA Question: 5



For the picture shown above, which one of the following is the correct picture representing reflection with respect to the mirror shown as the dotted line?



gatecivil-2022-set2 spatial-aptitude mirror-image

Answer key

10.5.6 Mirror Image: GATE Civil 2023 Set 1 | GA Question: 5



A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.

The figure below consists of 20 unit squares arranged as shown. In addition to the given black squares, upto 5 more may be coloured black. Which one among the following options depicts the minimum number of boxes that must be coloured black to achieve two lines of symmetry? (The figure is representative)

a			b
	c	d	
e	f	g	h
	i		
j			k

A. d

B. c, d, i

C. c, i

D. c, d, i, f, g

gatecivil-2023-set1 spatial-aptitude mirror-image

Answer key

10.5.7 Mirror Image: GATE ECE 2024 | GA Question: 9



Examples of mirror and water reflections are shown in the figures below:



An object appears as the following image after first reflecting in a mirror and then reflecting on water.

5.24

The original object is

A.

B.

C. H5.2

gateece-2024 spatial-aptitude mirror-image

Answer key

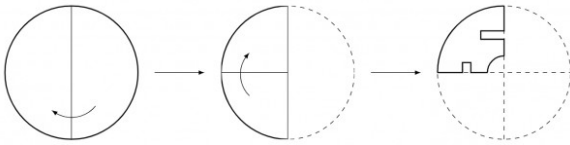
D. h.25

D.

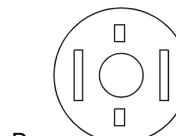
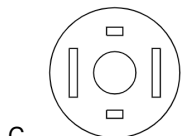
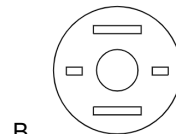
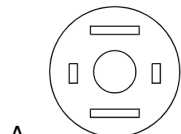
10.6

Paper Folding (12)

10.6.1 Paper Folding: GATE CSE 2021 Set 1 | GA Question: 4



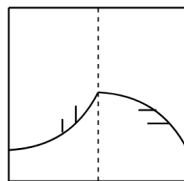
A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.



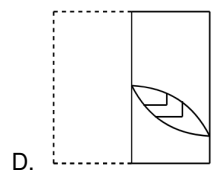
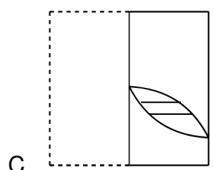
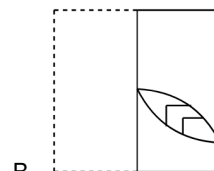
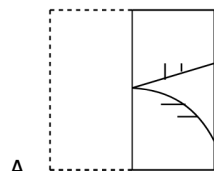
gatecse-2021-set1 spatial-aptitude paper-folding one-mark

Answer key

10.6.2 Paper Folding: GATE CSE 2021 Set 2 | GA Question: 2



A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.



gatecse-2021-set2 spatial-aptitude paper-folding one-mark

Answer key

10.6.3 Paper Folding: GATE CSE 2024 | Set 1 | GA Question: 9



A rectangular paper of $20\text{ cm} \times 8\text{ cm}$ is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

- A. 18 B. 24 C. 20 D. 21

gatecse2024-set1 spatial-aptitude paper-folding two-marks

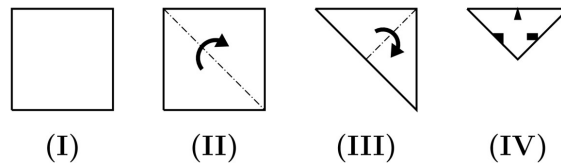
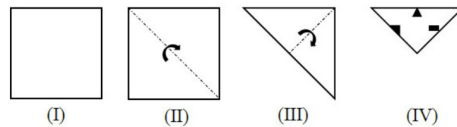
Answer key

10.6.4 Paper Folding: GATE CSE 2025 | Set 1 | GA Question: 9



A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

Note: The figures shown are representative.



- A.
- B.
- C.
- D.

- A.
- B.
- C.
- D.

gatecse2025-set1 spatial-aptitude paper-folding two-marks

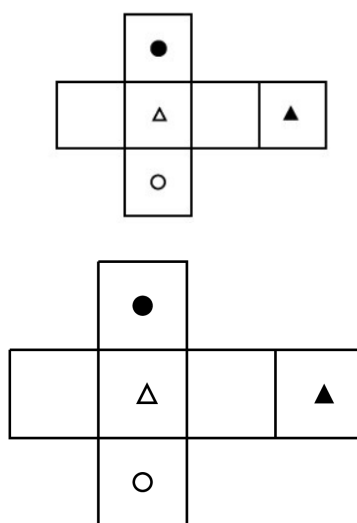
Answer key

10.6.5 Paper Folding: GATE CSE 2025 | Set 2 | GA Question: 4

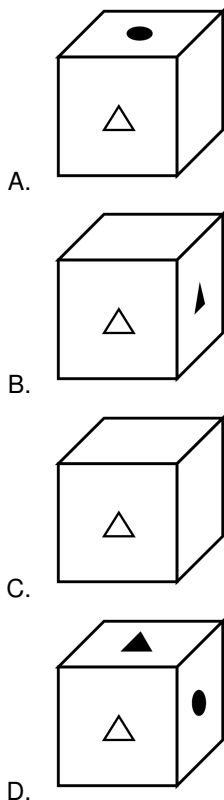
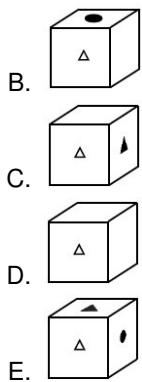


The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.



A.



gatecse2025-set2 spatial-aptitude paper-folding easy one-mark

Answer key

10.6.6 Paper Folding: GATE Civil 2021 Set 1 | GA Question: 7



Consider two rectangular sheets, Sheet M and Sheet N of dimensions $6\text{ cm} \times 4\text{ cm}$ each.

- Folding operation 1 : The sheet is folded into half by joining the short edges of the current shape.
- Folding operation 2 : The sheet is folded into half by joining the long edges of the current shape.

Folding operation 1 is carried out on Sheet M three times.

Folding operation 2 is carried out on Sheet N three times.

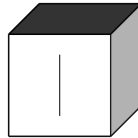
The ratio of perimeters of the final folded shape of Sheet N to the final folded shape of Sheet M is _____.

- A. 13 : 7 B. 3 : 2 C. 7 : 5 D. 5 : 13

gatecivil-2021-set1 spatial-aptitude paper-folding

Answer key

10.6.7 Paper Folding: GATE Civil 2022 Set 1 | GA Question: 10



Consider a cube made by folding a single sheet of paper of appropriate shape. The interior faces of the cube are all blank. However, the exterior faces that are not visible in the above view may not be blank.

Which one of the following represents a possible unfolding of the cube?

- A.
- B.
- C.
- D.

gatecivil-2022-set1 spatial-aptitude paper-folding

Answer key

10.6.8 Paper Folding: GATE ECE 2023 | GA Question: 10



A $100 \text{ cm} \times 32 \text{ cm}$ rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions $100 \text{ cm} \times 1 \text{ cm}$.

The total number of creases visible when the sheet is unfolded is _____.

- A. 32 B. 5 C. 31 D. 63

gateece-2023 spatial-aptitude paper-folding

Answer key

10.6.9 Paper Folding: GATE ECE 2024 | GA Question: 10



Two identical sheets A and B, of dimensions $24 \text{ cm} \times 16 \text{ cm}$, can be folded into half using two distinct operations, FO1 or FO2.

In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge.

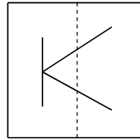
If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

- A. 14 : 11 B. 11 : 14 C. 18 : 11 D. 11 : 18

gateece-2024 geometry paper-folding spatial-aptitude

Answer key

10.6.10 Paper Folding: GATE Electrical 2021 | GA Question: 2



A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

- A.
- B.
- C.
- D.

gateee-2021 spatial-aptitude paper-folding

Answer key

10.6.11 Paper Folding: GATE Mechanical 2021 Set 2 | GA Question: 9



Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. The area of each face of the final folded shape, in square units, equal to _____.

- A. $\frac{1}{4}$ B. $\frac{1}{8}$ C. $\frac{1}{16}$ D. $\frac{1}{32}$

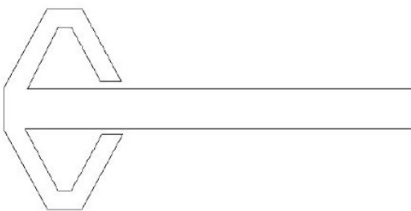
gateme-2021-set2 spatial-aptitude paper-folding area

Answer key

10.6.12 Paper Folding: GATE Mechanical 2024 | GA Question: 9



A planar rectangular paper has two V-shaped pieces attached as shown below.



This piece of paper is folded to make the following closed three-dimensional object.



The number of folds required to form the above object is

- A. 9 B. 7 C. 11 D. 8

gateme-2024 spatial-aptitude paper-folding

Answer key

10.7.1 Patterns In Three Dimensions: GATE CH 2023 | GA Question: 10



Which one of the following shapes can be used to tile (completely cover by repeating) a flat plane, extending to infinity in all directions, without leaving any empty spaces in between them? The copies of the shape used to tile are identical and are not allowed to overlap.

- A. circle
 B. regular octagon
 C. regular pentagon
 D. rhombus

gatech-2023 spatial-aptitude patterns-in-three-dimensions

Answer key

10.7.2 Patterns In Three Dimensions: GATE CSE 2023 | GA Question: 5



Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?

- A. The surface of the object must be concave everywhere.
 B. The surface of the object must be convex everywhere.
 C. The surface of the object may be concave in some places and convex in other places.
 D. The object can have edges, but no corners.

gatecse-2023 spatial-aptitude patterns-in-three-dimensions one-mark

Answer key

10.7.3 Patterns In Three Dimensions: GATE CSE 2024 | Set 2 | GA Question: 9



A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

- A. 3
 B. 4
 C. 7
 D. 8

gatecse2024-set2 spatial-aptitude patterns-in-three-dimensions two-marks

Answer key

10.7.4 Patterns In Three Dimensions: GATE Civil 2024 Set 2 | GA Question: 9



Five cubes of identical size and another smaller cube are assembled as shown in Figure A. If viewed from direction X, the planar image of the assembly appears as Figure B.

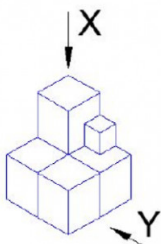
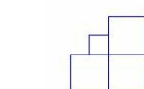
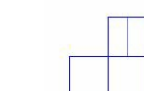
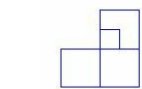
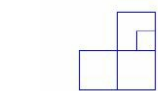


Figure A



Figure B

If viewed from direction Y, the planar image of the assembly (Figure A) will appear as



Answer key

10.7.5 Patterns In Three Dimensions: GATE DS&AI 2024 | GA Question: 10



Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

- A. A rhombus B. A triangle C. An ellipse D. A hexagon

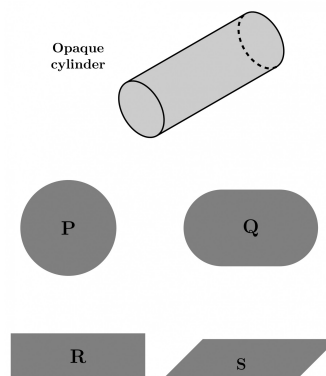
gate-ds-ai-2024 spatial-aptitude patterns-in-three-dimensions two-marks

Answer key

10.7.6 Patterns In Three Dimensions: GATE IN 2023 | GA Question: 10



An opaque cylinder (shown below) is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The cylinder can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P**, **Q**, **R**, and **S** is NOT possible?



- A. P B. Q C. R D. S

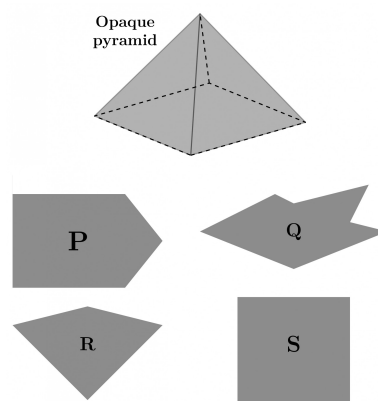
gatein-2023 spatial-aptitude patterns-in-three-dimensions

Answer key

10.7.7 Patterns In Three Dimensions: GATE Mechanical 2023 | GA Question: 10



An opaque pyramid (shown below), with a square base and isosceles faces, is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The pyramid can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P**, **Q**, **R**, and **S** is NOT possible?



- A. P B. Q C. R D. S

gateme-2023 spatial-aptitude patterns-in-three-dimensions

Answer key

10.7.8 Patterns In Three Dimensions: GATE Mechanical 2024 | GA Question: 10



Four equilateral triangles are used to form a regular closed three-dimensional object by joining along the edges. The angle between any two faces is

- A. 30° B. 60° C. 45° D. 90°

gateme-2024 spatial-aptitude patterns-in-three-dimensions

Answer key

10.8 Patterns In Two Dimensions (9)

10.8.1 Patterns In Two Dimensions: GATE CSE 2021 Set 1 | GA Question: 2



A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.

Which one of the following is NOT a convex polygon?

- A.  B.  C.  D. 

gatecse-2021-set1 spatial-aptitude patterns-in-two-dimensions one-mark

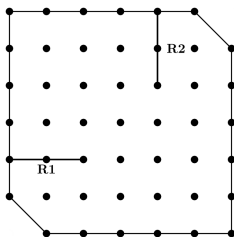
Answer key

10.8.2 Patterns In Two Dimensions: GATE CSE 2022 | GA Question: 10



A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved.

What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.

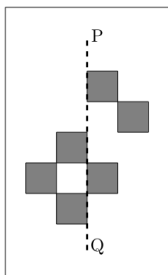


- A. 2 B. 4 C. 5 D. 3

gatecse-2022 spatial-aptitude patterns-in-two-dimensions two-marks

Answer key

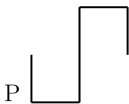
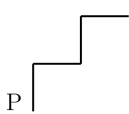
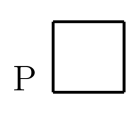
10.8.3 Patterns In Two Dimensions: GATE ECE 2021 | GA Question: 3



The least number of squares that must be added so that the line $P - Q$ becomes the line of symmetry is _____

- A. 4 B. 3 C. 6 D. 7

gateec-2021 spatial-aptitude patterns-in-two-dimensions

- B. 
- C. 
- D. 

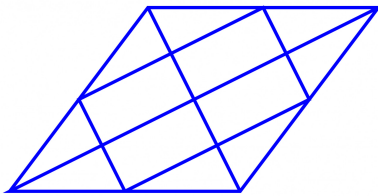
gate2022-ee spatial-aptitude patterns-in-two-dimensions

Answer key 

10.8.8 Patterns In Two Dimensions: GATE Electrical 2023 | GA Question: 5



How many triangles are present in the given figure?

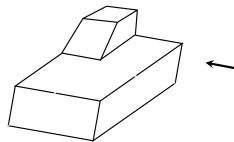


- A. 12 B. 16 C. 20 D. 24

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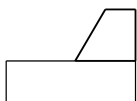
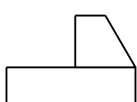
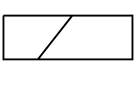
Answer key 

10.8.9 Patterns In Two Dimensions: GATE Mechanical 2022 Set 1 | GA Question: 5



A block with a trapezoidal cross-section is placed over a block with rectangular cross section as shown above.

Which one of the following is the correct drawing of the view of the 3D object as viewed in the direction indicated by an arrow in the above figure?

- A. 
- B. 
- C. 
- D. 

gateme-2022-set1 spatial-aptitude patterns-in-two-dimensions

Answer key 

Answer Keys

10.1.1	C
10.3.1	C
10.4.5	A
10.5.3	B

10.1.2	A
10.4.1	B
10.4.6	A
10.5.4	B

10.1.3	A
10.4.2	B
10.4.7	B
10.5.5	A

10.1.4	B,C
10.4.3	A
10.5.1	C
10.5.6	B

10.2.1	B
10.4.4	A
10.5.2	A
10.5.7	A

10.6.1	A
10.6.6	A
10.6.11	B
10.7.4	A
10.8.1	A
10.8.6	D

10.6.2	B
10.6.7	X
10.6.12	A
10.7.5	A
10.8.2	D
10.8.7	C

10.6.3	A
10.6.8	C
10.7.1	D
10.7.6	D
10.8.3	C
10.8.8	C

10.6.4	A
10.6.9	A
10.7.2	C
10.7.7	B
10.8.4	C
10.8.9	A

10.6.5	A
10.6.10	C
10.7.3	A
10.7.8	X
10.8.5	C